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SPPU M.Sc. Data Science — Free Study Material

Fundamentals of Data Science

DS-503-MJ

Sample / Practice Question Paper

Semester I · Core · Practice Paper

& This is a SAMPLE / PRACTICE question paper prepared for revision from the SPPU M.Sc. Data Science syllabus. It is NOT an official or past university examination paper. Use it only to test your preparation.

Time: 3 Hours

Maximum Marks: 35

Instructions: (1) All questions carry equal weightage where indicated. (2) Figures to the right indicate full marks. (3) Draw neat diagrams / write code wherever necessary.

Q1. Answer any FIVE of the following. [5 × 2 = 10]

- a) State the 3 V's of Big Data.
- b) What is structured data?
- c) Define an outlier.
- d) What is data wrangling?
- e) Name two data visualization tools.
- f) What is one-hot encoding?
- g) Define the data science lifecycle.

Q2. Answer the following. [10]

A) Explain the data science lifecycle and the data scientist's toolbox.
OR

B) Explain types of data and problems with unstructured data.

Q3. Answer the following. [10]

A) Explain data preprocessing: cleaning, transformation and encoding techniques.
OR

B) Explain proximity measures (Euclidean, Manhattan, Minkowski) and outlier detection.

Q4. Answer the following. [5]

A) Explain basic and specialized data visualization tools with use cases.
OR

B) Explain descriptive vs inferential statistics in data science.

